

Task – 5
Core Agent Development

Summary of Milestone-2

What I Built

In this milestone, I created a **complete multi-agent research system** using Python, LangGraph, and OpenAI/Tavily APIs.

Instead of writing one big script like before, I developed a **modular structure** with three separate agents:

1. Planner Agent

- Takes the main question.
- Breaks it into smaller sub-questions.
- Returns a clean Python list.

2. Searcher Agent

- Uses Tavily API.
- Searches the web for each sub-question.
- Returns short, structured summaries.

3. Writer Agent

- Combines planner instructions + search results.
- Generates the final research answer.

4. LangGraph Pipeline

I connected all agents using a **state-based workflow**:

User → Planner → Searcher → Writer → Output

What I Learned

a. Multi-Agent Architecture

I learned how real AI systems split tasks into small independent steps, instead of writing one big model call.

b. Working with State Models

I understood how LangGraph's StateGraph passes data between agents using ResearchState.

c. How Agent Pipelines Work

I learned how an entire research loop works:

1. Input →
2. Planning →
3. Searching →
4. Writing →
5. Final Output

d. Using API Keys & Env Variables

I learned how:

- OpenAI keys work
- Tavily keys work
- Environment variables are loaded

e. Debugging Real Errors

I experienced real-world issues like:

- Missing modules
- Invalid keys
- Rate limit errors

Advantages

1.Modular Design Like Real AI Systems

2.Scalable & Extensible

3.Clean Separation of Roles

Planner thinks → Searcher fetches → Writer summarizes.

4.Easy Debugging

If something fails,

I know exactly:

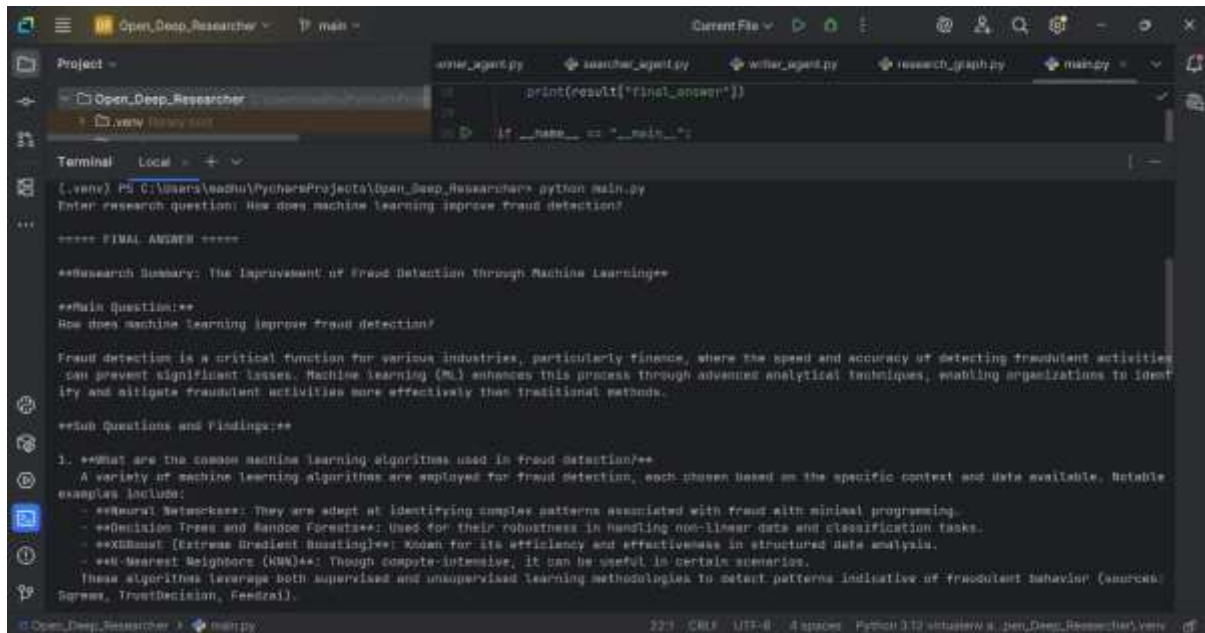
- Which agent broke
- Which step of the pipeline
- What part of the state caused the error

Limitations I Observed

Rate Limits

Tavily API Key Issues

Output Screenshots :



```
Project = Open_Deep_Researcher
main.py

Current File = main.py
writer_agent.py
searcher_agent.py
writer_agent.py
research_graph.py
main.py

print(result["final_answer"])

if __name__ == "__main__":

[...]\PS C:\Users\adhu\P\chess\Projects\Open_Deep_Researcher> python main.py
Enter research question: How does machine learning improve fraud detection?

**** FINAL ANSWER ****

**Research Summary: The Improvement of Fraud Detection through Machine Learning**

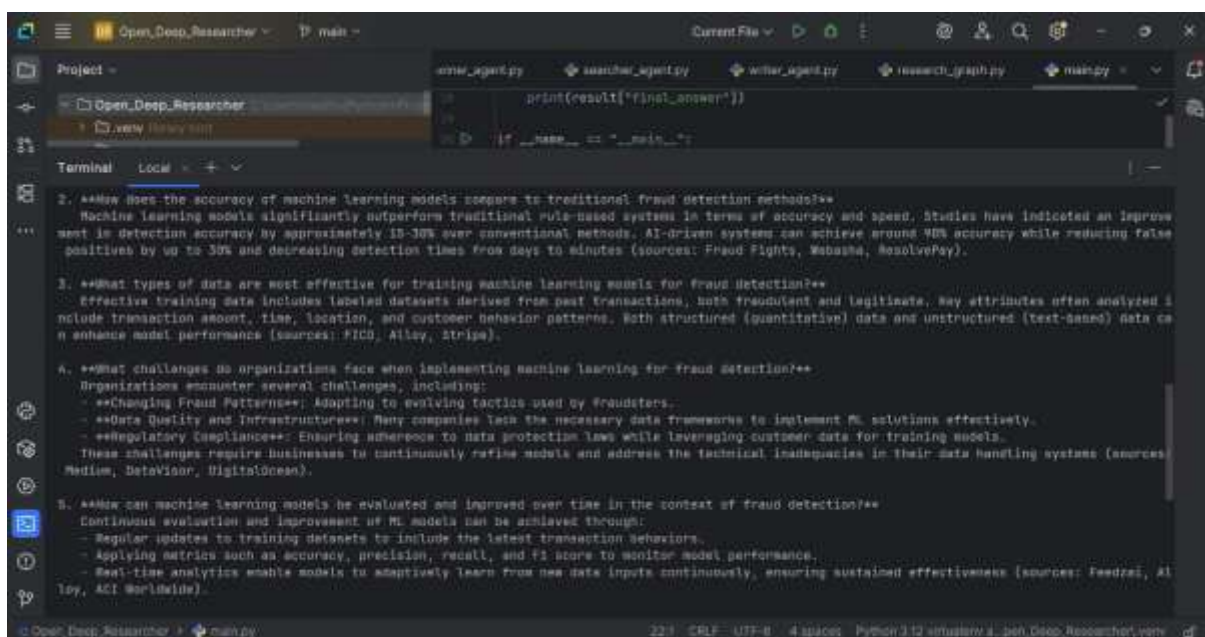
**Main Question:**
How does machine learning improve fraud detection?

Fraud detection is a critical function for various industries, particularly finance, where the speed and accuracy of detecting fraudulent activities can prevent significant losses. Machine learning (ML) enhances this process through advanced analytical techniques, enabling organizations to identify and mitigate fraudulent activities more effectively than traditional methods.

**Sub Questions and Findings:**

1. **What are the common machine learning algorithms used in fraud detection?**
A variety of machine learning algorithms are employed for fraud detection, each chosen based on the specific context and data available. Notable examples include:
- **Neural Networks:** They are adept at identifying complex patterns associated with fraud with minimal programming.
- **Decision Trees and Random Forests:** Used for their robustness in handling non-linear data and classification tasks.
- **XGBoost (Extreme Gradient Boosting):** Known for its efficiency and effectiveness in structured data analysis.
- **k-Nearest Neighbors (KNN):** Though compute-intensive, it can be useful in certain scenarios.
These algorithms leverage both supervised and unsupervised learning methodologies to detect patterns indicative of fraudulent behavior (sources: Scrum, TrustDecision, Feedzai).

[...]
```



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**Research Summary: The Improvement of Fraud Detection through Machine Learning**

**Main Question:**
How does machine learning improve fraud detection?

Fraud detection is a critical function for various industries, particularly finance, where the speed and accuracy of detecting fraudulent activities can prevent significant losses. Machine learning (ML) enhances this process through advanced analytical techniques, enabling organizations to identify and mitigate fraudulent activities more effectively than traditional methods.

**Sub Questions and Findings:**

2. **How does the accuracy of machine learning models compare to traditional fraud detection methods?**
Machine learning models significantly outperform traditional rule-based systems in terms of accuracy and speed. Studies have indicated an improvement in detection accuracy by approximately 15-30% over conventional methods. AI-driven systems can achieve around 90% accuracy while reducing false positives by up to 30% and decreasing detection times from days to minutes (sources: Fraud Fights, Webosha, ResolvePay).

3. **What types of data are most effective for training machine learning models for fraud detection?**
Effective training data includes labeled datasets derived from past transactions, both fraudulent and legitimate. Key attributes often analyzed include transaction amount, time, location, and customer behavior patterns. Both structured (quantitative) data and unstructured (text-based) data can enhance model performance (sources: FICO, Alloy, Stripe).

4. **What challenges do organizations face when implementing machine learning for fraud detection?**
Organizations encounter several challenges, including:
- **Changing Fraud Patterns:** Adapting to evolving tactics used by fraudsters.
- **Data Quality and Infrastructure:** Many companies lack the necessary data frameworks to implement ML solutions effectively.
- **Regulatory Compliance:** Ensuring adherence to data protection laws while leveraging customer data for training models.
These challenges require businesses to continuously refine models and address the technical inadequacies in their data handling systems (sources: Medium, DataVision, DigitsOcean).

5. **How can machine learning models be evaluated and improved over time in the context of fraud detection?**
Continuous evaluation and improvement of ML models can be achieved through:
- Regular updates to training datasets to include the latest transaction behaviors.
- Applying metrics such as accuracy, precision, recall, and F1 score to monitor model performance.
- Real-time analytics enable models to adaptively learn from new data inputs continuously, ensuring sustained effectiveness (sources: Feedzai, Alloy, ACI Worldwide).

[...]
```

```
Project - Open_Deep_Researcher
- venv
- main.py

Terminal Local
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**Conclusion:**
Machine learning significantly enhances the fraud detection landscape by employing sophisticated algorithms that adapt to changing fraud patterns, improving accuracy and reducing false positives compared to traditional methods. However, implementing these technologies presents challenges that organizations must address to fully leverage their potential. Continuous evaluation and data enhancement are vital for maintaining the efficacy of ML systems in combating fraud.

[.venv] PS C:\Users\eadu\PycharmProjects\Open_Deep_Researcher>

22/1 - CRLF UTF-8 4 spaces Python 3.12 virtualenv at .\venv\Open_Deep_Researcher\venv
```